Session 2: Crystal Structure

- **Vocabulary (English) - Part 1**
- 1.primitive cell
- 2. lattice constant
- 3. diamond
- 4. zincblende lattice structures
- 5.face-centered cubic (fcc) lattices
- 6.rock-salt lattice
- 7.wurtzite lattice structures
- 8.hexagonal close-packed lattices
- 9. Miller indices
- 10.basis vector
- 11.Wigner-Seitz cell
- 12.reciprocal lattice
- 13. the first Brillouin zone

Session 2: Crystal Structure

- **Vocabulary (w/Chinese) - Part 1**
- 1. primitive cell - 原胞
- 2. lattice constant - 晶格常数
- 3. diamond - 金刚石结构
- 4. zincblende lattice structures - 闪锌矿晶格结构
- 5. face-centered cubic (fcc) lattices - 面心立方结构
- 6. rock-salt lattice - 岩盐晶格结构
- 7. wurtzite lattice structures - 纤维锌矿晶格结构
- 8. hexagonal close-packed lattices - 六方密排晶格结构
- 9. Miller indices - 密勒指数
- 10. basis vector - 基矢
- 11. Wigner-Seitz cell - 维格纳—塞茨原胞
- 12. reciprocal lattice - 倒格子
- 13. the first Brillouin zone - 第一布里渊区

Session 2: Crystal Structure

- **Vocabulary (English) - Part 2**
- *14. body-centered cubic (bcc)*
- 15. ionic bond
- 16. covalent bond
- 17. metallic bond
- 18. Van der Waals bond
- 19. lattice vibration
- 20. point defect
- 21. Frenkel defect
- 22. line defect
- 23. line dislocation
- 24. substitutional impurities
- 25. interstitial impurities

Session 2: Crystal Structure

- **Vocabulary (w/Chinese) - Part 2**
- 14. body-centered cubic (bcc) - 体心立方
- 15. ionic bond - 离子键
- 16. covalent bond - 共价键
- 17. metallic bond - 金属键
- 18. Van der Waals bond - 范德华尔斯键
- 19. lattice vibration - 晶格振动
- 20. point defect - 点缺陷
- 21. Frenkel defect - 弗仑克尔缺陷
- 22. line defect - 线缺陷
- 23. line dislocation - 线位错
- 24. substitutional impurities - 替位杂质
- 25. interstitial impurities - 间隙杂质

Session 2: Crystal Structure

- **Vocabulary (English) - Part 3**
- 26. tetrahedral
- 27. interpenetrate
- 28. horizontally
- 29. sulfur
- 30. zinc sulfide
- 31. cadmium sulfide
- 32. orientation
- 33. breakage or cleavage plane
- 34. invariant
- 35. bisector
- 36. thermal equilibrium
- 37. inert element
- 38. repulsive
- 39. sodium chloride
- 40. dipole
- 41. gaseous
- 42. phosphorus
- 43. crystal

Session 2: Crystal Structure

- **Vocabulary (w/Chinese) - Part 3**

- 26. tetrahedral - 四面体的
- 27. interpenetrate - 相互贯通
- 28. horizontally - 水平地
- 29. sulfur - 硫
- 30. zinc sulfide - 硫化锌
- 31. cadmium sulfide - 硫化镉
- 32. orientation - 取向, 方位
- 33. breakage or cleavage plane - 解理面
- 34. invariant - 不变的
- 35. bisector - 平分线
- 36. thermal equilibrium - 热平衡
- 37. inert element - 惰性元素
- 38. repulsive - 排斥的
- 39. sodium chloride - 氯化钠
- 40. dipole - 偶极
- 41. gaseous - 气态的
- 42. phosphorus - 磷
- 43. crystal - 晶体

Sentences

- **1. The smallest assembly of atoms that can be repeated to form the entire crystal is called a primitive cell, with a dimension of lattice constant a.**
- *能够通过重复构建成完整晶体的最小原子组合被称为原胞，其尺寸为晶格常数a。*

Sentences

- **2. Many important semiconductors have diamond or zincblende lattice structures which belong to the tetrahedral phases.**
- *许多重要的半导体具有金刚石或闪锌矿晶格结构，它们都属于四面体相。*

Sentences

- **3. The diamond and the zincblende lattices can be considered as two interpenetrating face-centered cubic (fcc) lattices.**
- *金刚石和闪锌矿晶格可以被看作是两个相互贯通的面心立方(fcc)晶格。*

Sentences

- **4. Fig. 2.1 (f) shows the rock-salt lattice, which again can be considered as two interpenetrating face-centered cubic lattices.**
- *图2. 1(f)展示了岩盐晶格结构，它同样可以被视为两个相互贯穿的面心立方晶格。*

Sentences

- **5. Fig. 2.1(g) shows the wurtzite lattice, which can be considered as two interpenetrating hexagonal close-packed lattices.**
- *图2. 1(g)展示了纤维锌矿晶格，它可以被视为两个相互贯穿的六方密排晶格。*

Sentences

- **6. A convenient method of defining the various planes in a crystal is to use Miller indices.**
- *在晶体中定义不同晶面的一个便捷方法是使用密勒指数。*

Sentences

- **7. Three primitive basis vectors, a, b and c of a primitive cell, describe a crystalline solid such that the crystal structure remains invariant under translation.**
- *原胞的三个基本基矢a、b和c描述了一个晶体固体，使得该晶体结构在平移下保持不变。*

Sentences

- **8. The primitive cell of a reciprocal lattice can be represented by a Wigner-Seitz cell.**
- *倒格子的原胞可以用一个维格纳—塞茨原胞来表示。*

Sentences

- **9. The Wigner-Seitz cell in the reciprocal lattice is called the first Brillouin zone.**
- *倒格子中的维格纳—塞茨原胞被称为第一布里渊区。*

Sentences

- **10. Fig. 2.3(a) shows a typical example for a body-centered cubic (bcc) reciprocal lattice.**
- *图2.3(a)展示了一个典型的体心立方(bcc)倒格子的例子。*

Sentences

- **11. These oppositely charged ions then experience a coulomb attraction and form a bond referred to as an ionic bond.**
- *这些带相反电荷的离子会经历库仑吸引，并形成一种被称为离子键的化学键。*

Sentences

- **12. Another atomic bond that tends to achieve closed-valence energy shells is covalent bonding.**
- *另一种倾向于形成闭合价电子壳层的原子键是共价键。*

Sentences

- **13. The third major atomic bonding scheme is referred to as metallic bonding.**
- *第三种主要的原子键合方式被称为金属键。*

Sentences

- **14. A fourth type of atomic bond, called the Van der Waals bond, is the weakest of the chemical bonds.**
- *第四种被称为范德华尔斯键的原子键，是化学键中最弱的一种。*

Sentences

- **15. This imperfection, called lattice vibrations, affects some electrical parameters.**
- *这种被称为晶格振动的缺陷会影响一些电学参数。*

Sentences

- **16. Another type of defect is called a point defect.**
- *另一种类型的缺陷被称为点缺陷。*

Sentences

- **17. This vacancy-interstitial defect, also known as a Frenkel defect, produces different effects than the simple vacancy or interstitial.**
- *这种空位-填隙缺陷，也被称为弗仑克尔缺陷，其产生的影响不同于简单的空位或填隙缺陷。*

Sentences

- **18. A line defect, for example, occurs when an entire row of atoms is missing from its normal lattice site.**
- *例如，当一整排原子从其正常的晶格位置上缺失时，就会出现线缺陷。*

Sentences

- **19. This defect is referred to as a line dislocation.**
- *这种缺陷被称为线位错。*

Sentences

- **20. Impurity atoms may be located at normal lattice sites, in which case they are called substitutional impurities.**
- *杂质原子可能位于正常的晶格位置上，这种情况下它们被称为替位杂质。*

Sentences

- **21. Impurity atoms may also be located between normal sites, in which case they are called interstitial impurities.**
- *杂质原子也可能位于正常格点之间，此时它们被称为间隙杂质。*

Sentences

- **22. In this picture, for each sublattice (Cd or S), the two planes of adjacent layers are displaced horizontally.**
- *在此图中，对于每个子晶格（Cd或S），相邻层的两个晶面会水平地位移。*

Sentences

- **23. ...the sublattices of cadmium and sulfur.**
- *. . . 镉和硫的子晶格。*

Sentences

- **24. Some compounds, such as zinc sulfide and cadmium sulfide, can crystallize in either zincblende or wurtzite structures.**
- *一些化合物，例如硫化锌和硫化镉，既可以以闪锌矿结构结晶，也可以以纤维锌矿结构结晶。*

Sentences

- **25. Since semiconductor devices are built on or near the semiconductor surface, the orientations and properties of the surface crystal planes are important.**
- *由于半导体器件是制造在半导体表面上或其附近的，因此表面晶面的取向和性质非常重要。*

Sentences

- **26. For silicon, a single-element semiconductor, the easiest breakage or cleavage planes are the {111} planes.**
- *对于单元素半导体硅来说，最容易的断裂或解理面是{111}晶面。*

Sentences

- **27. Three primitive basis vectors...describe a crystalline solid such that the crystal structure remains invariant under translation.**
- *三个基本基矢. . . 描述了一个晶体固体，使得该晶体结构在平移下保持不变。*

Sentences

- **28. The Wigner-Seitz cell is constructed by drawing perpendicular bisector planes in the reciprocal lattice.**
- *维格纳—塞茨原胞的构建方法是在倒格子中绘制垂直平分面。*

Sentences

- **29. A fundamental law of nature is that the total energy of a system in thermal equilibrium tends to reach a minimum value.**
- *自然界的一条基本法则是，一个处于热平衡状态的系统的总能量趋向于达到最小值。*

Sentences

- **30. The atoms at the two extremes of the periodic table (excepting the inert elements) tend to lose or gain valence electrons.**
- *元素周期表两端的原子（惰性元素除外）倾向于失去或得到价电子。*

Sentences

- **31. If the ions were to get too close, a repulsive force would become dominant.**
- *如果离子之间距离太近，排斥力将占主导地位。*

Sentences

- **32. A classic example of ionic bonding is sodium chloride.**
- *离子键的一个经典例子是氯化钠。*

Sentences

- **33. This non-symmetry in the charge distribution results in a small electric dipole.**
- *电荷分布的这种非对称性导致了一个微小的电偶极。*

Sentences

- **34. ...most of these materials are in gaseous form at room temperature.**
- *. . . 这些材料大多数在室温下呈气态。*

Sentences

- **35. ...other impurities, such as gold or phosphorus in silicon, can drastically alter the electrical properties of the material.**
- *. . . 然而，其他杂质，例如硅中的金或磷，可以极大地改变材料的电学特性。*

Sentences

- **36. In a real crystal, the lattice is not perfect, but contains imperfections or defects.**
- *在真实的晶体中，晶格并非完美，而是包含瑕疵或缺陷。*

Sentences

- **37. In this rock-salt structure, each atom has six nearest neighbors.**
- *在这种岩盐结构中，每个原子有六个最近邻原子。*

Sentences

- **38. The interaction between atoms can he described by quantum mechanics.**
- *原子间的相互作用可以通过量子力学来描述。*

The end of Session 2.